## Sugar as a source of energy

Sugar is an important source of energy. During digestion, all carbohydrates in food (starches and sugars) break down into simple sugars. These molecular sugars are absorbed from the intestine into the blood stream and travel to the cells, where they are used to provide energy for cellular functions. In parts of the world, where people suffer from energy malnutrition and are undernourished, sugar is valued as an inexpensive source of energy to support human activities.

## 5.5.3 Stages of sugar cookery

When sugar is boiled at different temperatures, various sweets can be made.

## 5.5.4 Jaggery

Jaggery is a concentrated product of cane juice and can vary from golden brown to dark brown in colour. It contains upto 50% sucrose, 20% invert sugars and 20% moisture. It is mostly produced in India and Africa. It is used to make several Indian desserts and is known as a healthy replacement of sugar, primarily because it is unprocessed.

### Forms of Jaggery

- Jaggery square – Achu vellam
- Jaggery palm – *Karuppatti*
- Jaggery (paagu) – *Paagu vellam*

Jaggery can also provide therapeutic benefits, and hence used as various medicines in ayurveda and folk medicine.

**Table 5.2 Stages of sugar cookery**

| Product | Temperature | Test | Description of |
|---|---|---|---|
| Syrup | 110-112°C | Thread | When syrup is dropped from a spoon, syrup spins a 5cm thread |
| barfi, fondant,fudge | 112-115°C | Soft ball | Froms a soft ball when syrup is dropped in cold water. |
| Caramels | 118-120°C | Firm ball | Froms a firm ball when syrup is dropped in cold water, does not flatten on removal from water. |
| Divinity, laddu, marshmellow | 120-130°C | Hard ball | Froms a ball hardenough to hold its shape when syrup is dropped in cold water. |
| Butterscotcht | 132-143°C | Soft crack | Froms threads which are hard but not brittle when syrup is dropped in cold water. |
| Brittle | 150-154°C | Hard crack | Froms threads which are brittle when syrup is dropped. |
| Caramel | 170°C | brown liquid | Sugar melts and browns. |



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